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Data Structures and Algorithm

- Data structures provide different ways to organize and store data so that it can be accessed, observed, and managed efficiently. Algorithms are the step-by-step process that is used to perform tasks and solve problems. When both are utilized and worked together, they help program processes and data managing run efficiently.

Linear Data Structure

- Linear Data Structure are utilized to organize data in an order, whereas elements are arranged properly. It is useful when data needs to be processed in a sequence.

One-Dimension Array

- One-Dimension array is a type of data that linear structured and stores multiple elements in sequence. Each are easily identified by index.

The Big O Notation

- Big O notation is used to describe how an algorithm's memory requirement increase as the amount of input data grows, common examples are time. It is used to measure and compare algorithm efficiently.

Basic Searching Algorithms

- Search Algorithms are set of methods used to find a value within the collection of data. Two basic searching algorithms are linear and binary search. The use of each search depends on the factors such as the size and organization of the data.

Linear Search

- Linear Search checks each element one at a time until the target value has been reached. It does require the data to be sorted, it can be slower for large datas.



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Binary Search

- Whereas binary search is used to find value by repeatedly dividing a sorted collection, It checks the middle element and eliminate the half that cannot contain the target.

REFERENCES:

- [What is a Data Structure? | IBM](#)
- [Introduction to Linear Data Structures - GeeksforGeeks](#)
- [Array Data Structure - GeeksforGeeks](#)
- [Big O Notation - GeeksforGeeks](#)
- [Searching Algorithms - GeeksforGeeks](#)
- [What is Linear Search? - GeeksforGeeks](#)
- [Binary Search - GeeksforGeeks](#)